Serie 21

Diagonalizabiltiy, Cayley-Hamilton

1. Let K be a field. Let $p(X) \in K[X]$ be a polynomial. Let $A, Q \in M_{n\times n}(K)$ be matrices such that Q is invertible. Show that
$$p(Q^{-1}AQ) = Q^{-1}p(A)Q$$

2. For $k \in \mathbb{N}$, let $p_k(X) = \sum_{i=0} c_i^{(k)} X^i, p(X) = \sum_{i=0} c_i X^i \in \mathbb{C}[X]$ be polynomials such that $\deg(p_k) \leqslant d$ for some fixed positive integer d. We say that p_k converges to p, as $k \to \infty$ if
$$c_d^{(k)} X^d + c_{d-1}^{(k)} X^{d-1} + \cdots + c_0^{(k)}$$
converges to $c_d X^d + c_{d-1} X^{d-1} + \cdots + c_0^{(k)}$ as $k \to \infty$. If $A, A_k \in M_{n\times n}(\mathbb{C})$ are matrices such that $A_k \to A$ as $k \to \infty$ and p_k converges to p, then show
$$p_k(A_k) \to p(A)$$
as $k \to \infty$.

3. Let $\varepsilon > 0$ and $A = (a_{ij})_{i,j} \in M_{n\times n}(\mathbb{C})$. Show that there exists a matrix $B = (b_{ij})_{i,j} \in M_{n\times n}(\mathbb{C})$ which is upper diagonalisable, such that $|b_{ij} - a_{ij}| < \varepsilon$.

4. Using the exercises above, prove the Cayley-Hamiltton theorem.

   Hint: First show it for diagonal matrices, then use Exercise 1 to show it for diagonalisable matrices).